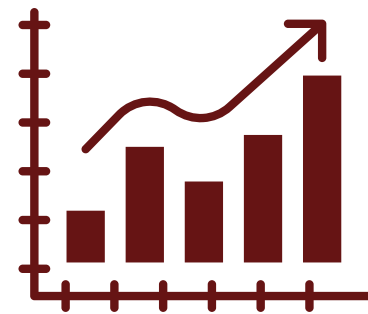


# Telco Customer Churn Analysis

Customer Retention Insights Using SQL & Power BI



Presented by

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*Tools:*  *Excel* |  *SQL* |  *Power BI*

# Project Overview

## Business Problem

Telecommunication companies face customer attrition, which impacts revenue and growth. Understanding why customers leave the company is essential for improving customer retention and reducing churn.

## Project Objective

The objective of this project is to analyze customer data using SQL and identify key factors contributing to customer churn. The analysis aims to uncover patterns related to contracts, tenure, services and monthly charges to support data-driven business decisions.

## Dataset Overview

The dataset contains customer demographic information, account details, subscribed services, billing information and churn status. It includes attributes such as gender, tenure, contract type, internet service, payment method, monthly charge, total charges and customer churn.

# **SQL Techniques used...**

- Aggregate Functions (COUNT, AVG)
- GROUP BY
- CASE WHEN
- Window Functions (RANK)
- Common Table Expressions (CTEs)
- Conditional Aggregation
- Churn Rate Calculations

# Overall churn Analysis

## SQL Query:

```
SELECT
    ROUND(100 * SUM(CASE
        WHEN churn = 'yes' THEN 1
        ELSE 0
    END) / COUNT(*),
    2) AS churn_rate
FROM
    churn;
```

## Result:

| Result Grid |            |
|-------------|------------|
|             | churn_rate |
| ▶           | 26.58      |

## Insight:

26.58% of customers have churned, indicating that approximately 1 in 4 customers leave the company

# Contract type Analysis

## SQL Query:

```
SELECT
    contract, COUNT(*) AS customer_count
FROM
    churn
WHERE
    churn = 'yes'
GROUP BY contract
ORDER BY customer_count DESC
LIMIT 1;
```

## Result:

| Result Grid     Filter Rows: <input type="text"/> |                |                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
|                                                                                                                                                                                                                         | contract       | customer_count |
| ▶                                                                                                                                                                                                                       | Month-to-month | 1655           |

## Insight:

Customers with month-to month contracts recorded the highest number of churn cases compared to customers on One-Year and Two-Year contracts. This suggests that customer without a long term commitment are more likely to leave.



# Payment Method Analysis

## SQL Query:

```
SELECT
    COUNT(*) AS customers, paymentmethod
FROM
    churn
WHERE
    CHURN = 'YES'
GROUP BY paymentmethod
ORDER BY customers DESC;
```

## Result:

| Result Grid |           | Filter Rows:              |
|-------------|-----------|---------------------------|
|             | customers | paymentmethod             |
|             | 1071      | Electronic check          |
|             | 308       | Mailed check              |
|             | 258       | Bank transfer (automatic) |
|             | 232       | Credit card (automatic)   |

## Insight:

Electronic Check users showed the highest churn among all payment methods. Customers using this payment option appear to be more likely to discontinue their services.

# Internet Service Analysis

## SQL Query:

```
SELECT
    internetervice, COUNT(*) AS customers
FROM
    churn
WHERE
    churn = 'yes'
GROUP BY internetervice
ORDER BY customers DESC
LIMIT 1;
```

## Result:

| Result Grid     Filter Rows: |                |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
|                                                                                                                                                                                                    | internetervice | customers |
| +                                                                                                                                                                                                  | Fiber optic    | 1297      |

## Insight:

Fiber Optic customers contributed the largest number of churned customers among all internet service categories.

# Average tenure of Churned Customers

## SQL Query:

```
SELECT
    churn, ROUND(AVG(tenure), 2) AS avg_ten
FROM
    churn
GROUP BY churn;
```

## Result:

| Result Grid |       |         |  |  |
|-------------|-------|---------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|             | churn | avg_ten |                                                                                     |                                                                                     |
|             | No    | 37.65   |                                                                                     |                                                                                     |
|             | Yes   | 17.98   |                                                                                     |                                                                                     |

## Insight:

Customers who churned stayed with the company for an average of only 17.98 months, while retained customers stayed much longer. This shows that newer customers are more likely to leave.



# Average Monthly Charges Analysis

## SQL Query:

```
SELECT
    churn, ROUND(AVG(monthlycharges), 2) AS avg_mon_chrg
FROM
    churn
GROUP BY churn;
```

## Result:

| Result Grid     Filter |       |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------|
|                                                                                                                                                                                              | churn | avg_mon_chrg |
| ▶                                                                                                                                                                                            | No    | 61.31        |
|                                                                                                                                                                                              | Yes   | 74.44        |

## Insight:

Churned customers had an average monthly charge of 74.44 compared to 61.31 for retained customers. Customers paying higher charges are more likely to leave.

# Tenure Segmentation Analysis

## SQL Query:

```
SELECT
  CASE
    WHEN tenure < 12 THEN 'New'
    WHEN tenure BETWEEN 12 AND 48 THEN 'Regular'
    ELSE 'Loyal'
  END AS customer_segment,
  ROUND(100 * SUM(CASE
    WHEN churn = 'yes' THEN 1
    ELSE 0
  END) / COUNT(*),
  2) AS churn_rate
FROM
  churn
GROUP BY customer_segment;
```

## Result:

| Result Grid     Filter Rows: <input type="text"/> |                  |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------|
|                                                                                                                                                                                                                         | customer_segment | churn_rate |
| ▶                                                                                                                                                                                                                       | New              | 48.54      |
|                                                                                                                                                                                                                         | Regular          | 24.02      |
|                                                                                                                                                                                                                         | Loyal            | 9.51       |

## Insight:

Customer churn is highest among new customers and decrease as customer tenure increases. Long-term customers show much lower churn rates than recently acquired customers.

# Customer Ranking by Total Charges

## SQL Query:

```
SELECT customerID,  
       totalcharges,  
       rank()over(  
         order by totalcharges desc  
       )as rnk  
  
from  
  churn;
```

## Result:

| Result Grid    Filter Rows: <input type="text"/> |            |              |     |
|-------------------------------------------------------------------------------------------------------------------------------------|------------|--------------|-----|
|                                                                                                                                     | customerID | totalcharges | rnk |
|                                                                                                                                     | 2889-FPWRM | 8684.80      | 1   |
|                                                                                                                                     | 7569-NMZYQ | 8672.45      | 2   |
|                                                                                                                                     | 9739-JLPQJ | 8670.10      | 3   |
|                                                                                                                                     | 9788-HNGUT | 8594.40      | 4   |
|                                                                                                                                     | 8879-XUAHX | 8564.75      | 5   |

## Insight:

Ranking customers by Total Charges helps identify high-value customers who contribute significantly to overall revenue.

# Key Findings.....

- The overall churn rate is **26.58%**.
- Customers on **Month-to-Month** contracts showed the highest churn rate of **1655** compared to one year and two year contract customers.
- **Electronic Check** was the payment method with the highest churn, contributing **1071** churned customers.
- **Fiber Optic** Internet customers showed the highest churn among internet service categories contributing **1297** customers.
- Customers who churn have an average tenure of **17.98** months compared to **37.65** months for retained customers.
- Churned Customers have a higher average monthly charge of **74.44** compared to retained customers **61.31**.
- Tenure strongly impacts churn- new customers have the highest churn rate (**48.54%**), while loyal customers have the lowest (**9.51%**).

# Business Recommendations

- Encourage customers to switch from month-to-month plans to long-term contracts.
- Offer special onboarding and retention programs for new customers.
- Encourage customers to switch to more stable payment methods.
- Review pricing strategies for customers with high monthly charges.
- Improve customer experience for Fiber Optic Internet users.
- Retention efforts should focus on the first year to improve customer loyalty and reduce churn.